

KCMamba / KCA-Mamba block

$$x \in \mathbb{R}^{B \times T \times d}$$

in_proj ($d \rightarrow 2E$)

DepthwiseConv1d + SiLU

$v = \text{LN}(W_v x_{\text{core}})$

$h_t = A h_{t-1} + K v_t$ (prefix-scan)

$\mu \odot g$ (gating)

out_proj ($E \rightarrow d$) + residual

$$y \in \mathbb{R}^{B \times T \times d}$$

Param net: Q_t, R_t, K_Δ

$K_{\text{base}} = Q/(Q + R)$

$K = \text{clamp}(K_{\text{base}} + K_\Delta, 10^{-4}, 0.999)$

$A = \text{clamp}(1 - K, 0.01, 0.99)$

